

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)
ANSWER KEY
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	D	C	D	A	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	B,C,D	B,D	A,D	B,C,D	C,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.66 to 0.67	0.00	1.00	1.00	40.00	7.71

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	B	C	A	C	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,D	B,C,D	B,C,D	A,C	A,B,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.00	5.00	40.00	53.25	6.00	3.00

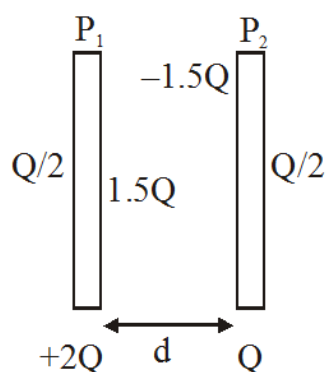
PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	C	D	C	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B	A,C,D	A,B,C,D	A,B,C,D	A,B	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	48.00	2.66 or 2.67	0.50	1560.00	8.00	4.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (B)

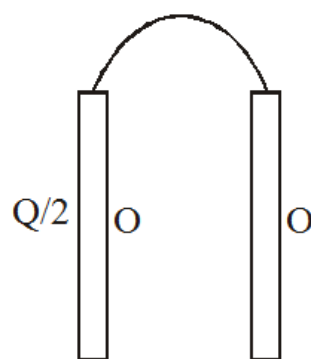
$$C = \frac{\epsilon_0 A}{d}, U_i = \frac{(1.5Q)^2}{2C} = \frac{2.25Q^2 d}{2\epsilon_0 A}$$



$$\Rightarrow U_i = \frac{9Q^2 d}{8\epsilon_0 A}$$

After connecting

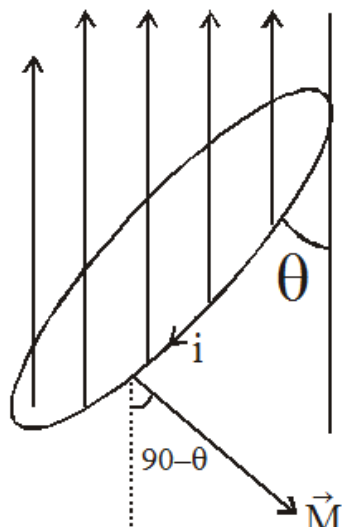
$$U_f = 0$$



$$\text{Heat} = U_i - U_f = \frac{9Q^2 d}{8\epsilon_0 A}$$

3. Ans (C)

At an angular position θ ,



$$\phi = \pi a^2 \sin \theta B$$

$$\varepsilon \approx \pi a^2 B \Omega$$

$$i = \frac{\pi a^2 B \Omega}{R}$$

$$\tau = \vec{M} \times \vec{B} = i \pi a^2 B \sin(90 + \theta)$$

$$= \frac{\pi^2 a^4 B^2 \Omega}{R}$$

$$\tau_{\text{ext}} = mg \sin \theta a \approx mg \theta a$$

$$\tau_{\text{ext}} = I \alpha$$

$$I \frac{d^2 \theta}{dt^2} + \frac{\pi^2 a^4 B^2 \Omega}{R} + mg a \theta = 0$$

$$\frac{d^2 \theta}{dt^2} + \frac{\pi^2 a^4 B^2 \Omega}{\frac{3}{2} m a^2 R} + \frac{mg a}{\frac{3}{2} m a^2} \theta = 0$$

$$\frac{d^2 \theta}{dt^2} + 2r\omega + \omega_0^2 \theta = 0$$

$$\text{Angular frequency} = \sqrt{\omega_0^2 - r^2}$$

$$= \sqrt{\frac{2g}{3a} - \frac{\pi^4 a^4 B^4}{9m^2 R^2}}$$

4. Ans (D)

$$\epsilon_0 E A = Q = \frac{\epsilon_0 A V}{d}$$

$$\Rightarrow v = Ed$$

$$i_{\text{max}} = Q_w = \frac{1}{\sqrt{LC}} \times CEd = Ed \sqrt{\frac{C}{L}}$$

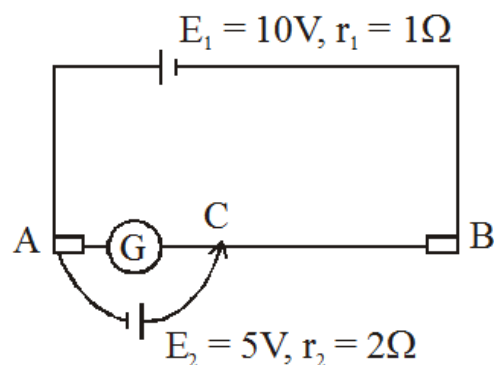
5. Ans (A)

Potential difference across AC is zero as

$$I_{AC} = 0$$

$$5 - 2I = 0$$

$$I = 2.5 \text{ A}$$



Let the resistance of part BC be r

Applying KVL

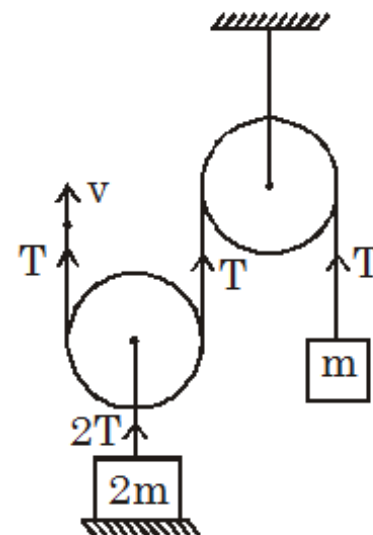
$$10 + 5 - 2I - Ir - I = 0$$

$$2.5r = 7.5 \Rightarrow r = 3 \Omega$$

As resistance of part AB = 9Ω

$$\therefore \text{Length AC} = 66.7 \text{ cm}$$

6. Ans (C)



$$\int 2T dt = 2mv_2$$

$$\int T dt = mv_1$$

$$\Rightarrow v_2 = v_1$$

$$Ts_1 + 2Ts_2 - Ts = 0$$

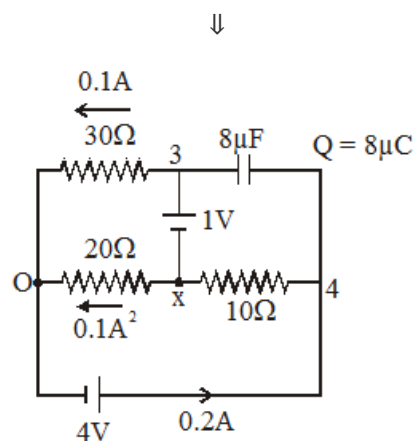
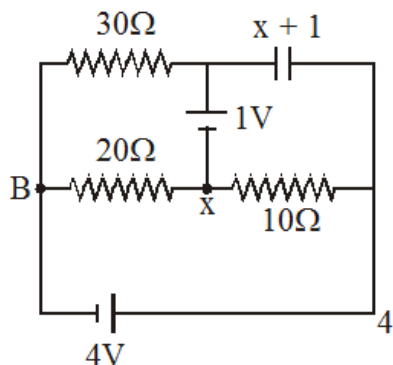
$$\Rightarrow s = s_1 + 2s_2 \Rightarrow v = 3v_1 \Rightarrow v_1 = \frac{v}{3}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B,C,D)

In steady state,



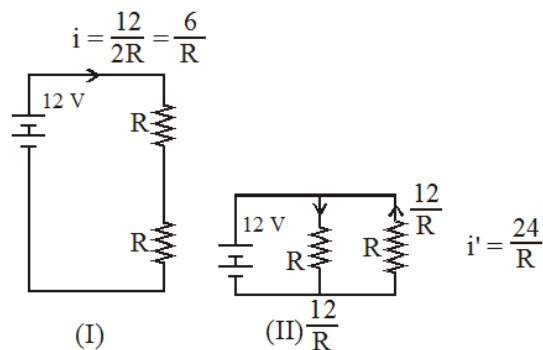
$$\frac{x+1-0}{30} + \frac{x-0}{20} + \frac{x-4}{10} = 0$$

$$\frac{2x+2+3x+6x-24}{6} = 0$$

$$11x = 22$$

$$x = 2\text{V}$$

8. Ans (B,D)



11. Ans (C,D)

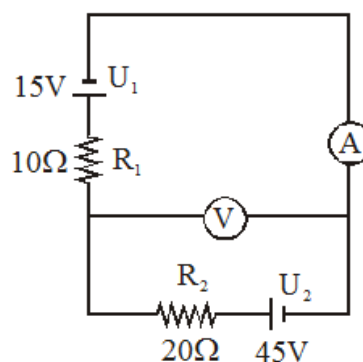
$$i = \frac{dQ}{dt} = 4e^{-0.2t} \times 4\pi \cos\left(4\pi t + \frac{\pi}{3}\right) + \sin\left(4\pi t + \frac{\pi}{3}\right) \times 4x - 0.2e^{-0.2t}$$

$$= 4e^{-0.2t} \left[4\pi \cos\left(4\pi t + \frac{\pi}{3}\right) - 0.8 \sin\left(4\pi t + \frac{\pi}{3}\right) \right]$$

$$\frac{di}{dt} = -0.8e^{-0.2t} \left[4\pi \left(\cos 4\pi t + \frac{\pi}{3} \right) - 0.8 \sin\left(4\pi t + \frac{\pi}{3}\right) \right]$$

$$+ 4e^{-0.2t} \left[-16\pi^2 \sin\left(4\pi t + \frac{\pi}{3}\right) - 3.2 \cos\left(4\pi t + \frac{\pi}{3}\right) \right]$$

12. Ans (B,C)



$$i = \frac{30}{30} = 1\text{A}$$

$$v = 45 - ir = 25$$

$$i' = \frac{u_2}{R_2} = \frac{45}{20} = 2.25\text{A}$$

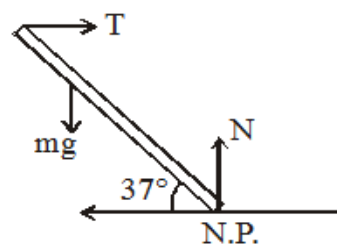
$$v' = u_1 = 15\text{V}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (0.66 to 0.67)

The FBD of any one rod is



$$T = \mu N \quad \dots(i)$$

$$mg = N \quad \dots(ii)$$

Taking torque of any one rod is

$$mg \frac{L}{g} \cos 37^\circ = TL \sin 37^\circ$$

$$T = \frac{mg}{2} \frac{4}{3} = \frac{2mg}{3}$$

$$\frac{2mg}{3} = \mu mg \Rightarrow \mu = \frac{2}{3}$$

2. Ans (0.00)

$$\omega_0^2 = \frac{g}{\ell} = 10$$

$$r = \frac{b\ell}{2_{\max}} = 1$$

$$\omega = \sqrt{\omega_0^2 - r^2}$$

$$= \sqrt{10 - 1} = 3 \text{ rad/s}$$

$$A = A_0 e^{-\tau t}$$

$$\frac{A_0}{2} = A_0 e^{-\tau t}$$

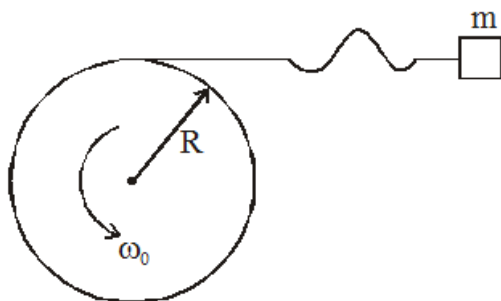
$$\ell n 2 = \tau t$$

$$t = 0.693$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{3}$$

$$t \text{ from 1 extreme to another} = \frac{T}{2} = \frac{\pi}{3}$$

3. Ans (1.00)



Angular momentum conservation

$$\frac{1}{2} MR^2 \omega_0 = \left(\frac{MR^2}{2} + MR^2 \right) \omega'$$

$$\left(\frac{M}{M + 2m} \right) \omega_0 = \omega'$$

$$U = \omega' R = \left(\frac{M}{M + 2m} \right) \omega_0 R$$

4. Ans (1.00)

$$t = \frac{dE[B^2 \ell C + M]}{B \ell m g} = 1000 \text{ sec}$$

5. Ans (40.00)

$$\omega Q = i = \frac{V_0}{\omega L - \frac{1}{\omega C}}$$

$$Q = \frac{V_0}{\omega^2 L - \frac{1}{C}}$$

$$A = \frac{F_0}{\omega_f^2 m - k}$$

$$\omega_f^2 m - k = \frac{F_0}{A}$$

$$k = \omega_f^2 m - \frac{F_0}{A}$$

$$= 25 \times 4 - \frac{900}{0.15} = 40 \text{ N/m}$$

6. Ans (7.71)

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$$

$$a = 3.5 \text{ m/s}^2 = \frac{10 \times \frac{1}{2}}{1 + \frac{I}{mR^2}}$$

$$1 + \frac{I}{mR^2} = \frac{10}{7}$$

$$I = \frac{3}{7} mR^2 = \frac{3}{7} \times 0.5 \times 6$$

$$I = \frac{54}{7} \text{ kgcm}^2 = 7.71 \text{ kg cm}^2$$

PART-2 : CHEMISTRY

SECTION-I (i)

5. Ans (C)

PART-2 : CHEMISTRY

SECTION-II

4. Ans (53.25)

2g chlorine used 3g metal for formation of metal chloride. Then for chlorine weight 35.5 g weight of metal

$$= \frac{3 \times 35.5}{2} = 53.25$$

5. Ans (6.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

$$\left(\frac{x+1+\frac{1}{x-1}}{1-\frac{x+1}{x-1}} \right) = \tan \left(2 \tan^{-1} \frac{4}{3} \right) = \tan \left(\pi - \tan^{-1} \frac{24}{7} \right)$$

$$\Rightarrow x^2 = \frac{48}{7}$$

$$x = \pm 4\sqrt{\frac{3}{7}}$$

2. Ans (C)

$$\text{Given } f(x+5) \geq f(x) + 5 \quad \dots(1)$$

$$f(x+1) \leq f(x) + 1 \quad \dots(2)$$

from (1) and (2), we can write

$$\begin{aligned} f(x) + 5 &\leq f(x+5) \leq f(x+4) + 1 \\ &\leq f(x+3) + 2 \\ &\leq f(x+2) + 3 \\ &\leq f(x+1) + 4 \\ &\leq f(x) + 5 \end{aligned}$$

$\therefore$ equality exist everywhere

$$\therefore f(x+1) = f(x) + 1 \quad \therefore f(1) = 1$$

$$\Rightarrow f(2) = 2, f(3) = 3 \dots f(2021) = 2021$$

$$\text{Now } g(x) = f(x) + 5 - x \Rightarrow g(2021) = 5$$

3. Ans (D)

$$([x] + [y] + [z])([x] - [y])^2 + ([y] - [z])^2 + ([z] - [x])^2 = 8$$

possible only when $[x] + [y] + [z] = 4$

$$[x] = 2, [y] = 1, [z] = 1$$

$$\text{or } [x] = 1, [y] = 2, [z] = 1$$

$$[x] = 1, [y] = 1, [z] = 2$$

4. Ans (C)

$$\therefore (1 + \sqrt{2})^n = A_n + B_n \sqrt{2}$$

$$\Rightarrow (1 - \sqrt{2})^n = A_n - B_n \sqrt{2}$$

$$\text{on solving, } A_n = \frac{(1 + \sqrt{2})^n + (1 - \sqrt{2})^n}{2},$$

$$B_n = \frac{(1 + \sqrt{2})^n - (1 - \sqrt{2})^n}{2\sqrt{2}}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{A_n}{B_n} = \lim_{n \rightarrow \infty} \sqrt{2} \cdot \left(\frac{1 + \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^n}{1 - \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^n} \right) = \sqrt{2}$$

$$\therefore \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^n \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\& (A) \rightarrow \text{limit} = 1 \quad (C) \rightarrow \text{limit} = \sqrt{2}$$

$$(B) \rightarrow \text{limit} = 0$$

5. Ans (C)

Let

$$S = \sum_{n=1}^{99} \sqrt{10 + \sqrt{n}}, \quad T = \sum_{n=1}^{99} \sqrt{10 - \sqrt{n}}.$$

Note that $\sqrt{a+b+2\sqrt{ab}} = \sqrt{a} + \sqrt{b}$ for

positive real numbers a and b . In particular, for each

fixed positive integer n with $1 \leq n \leq 99$, choose $a > b > 0$

satisfying $a + b = 20$, $ab = n$, then a, b are roots of the

equations $x^2 - 20x + n = 0$ i.e.

$$a = 10 + \sqrt{100 - n}, \quad b = 10 - \sqrt{100 - n}$$

$$\text{so } \sqrt{2S} = \sum_{n=1}^{99} \sqrt{20 + 2\sqrt{n}} = \sum_{n=1}^{99} (\sqrt{10 + \sqrt{100 - n}} + \sqrt{10 - \sqrt{100 - n}})$$

$$= \sum_{n=1}^{99} (\sqrt{10 + \sqrt{n}} + \sqrt{10 - \sqrt{n}}) = S + T$$

$$\text{Thus, } (\sqrt{2} - 1)S = T \Rightarrow \frac{S}{T} = \frac{1}{\sqrt{2} - 1} = 1 + \sqrt{2}$$

6. Ans (B)

$$\lim_{h \rightarrow 0} \frac{(\log(2+h) - \log 2)3 \cdot (4^h - 1 - h)}{((8+h)^{1/3} - (4+3h)^{1/2}) \cdot \sin \pi(1+h)}$$

$$3 \cdot \log \left(1 + \frac{h}{2} \right) \cdot (4^h - 1 - h)$$

$$\lim_{h \rightarrow 0} \frac{2 \left[\left(1 + \frac{h}{8} \right)^{1/3} - \left(1 + \frac{3h}{4} \right)^{1/2} \right] (-\sin \pi h)}{3 \cdot \left(\log \frac{(1+h/2)}{h/2} \cdot \frac{h}{2} \right) \cdot (4^h - 1 - h)}$$

$$\lim_{h \rightarrow 0} \frac{2 \left[\left(1 + \frac{h}{24} + \dots \right) - \left(1 + \frac{3h}{8} \right) \right] \cdot \left(\frac{-\sin \pi h}{\pi h} \right) \pi h}{2 \left(\frac{h}{3} \right) (\pi h)} = \lim_{h \rightarrow 0} \frac{9}{4\pi} \left(\frac{4^h - 1 - h}{h} \right)$$

Using standard limit

$$\lim_{h \rightarrow 0} \frac{\left(\frac{3h}{2} \right) (4^h - 1 - h)}{2 \left(\frac{h}{3} \right) (\pi h)} = \lim_{h \rightarrow 0} \frac{9}{4\pi} \left(\frac{4^h - 1 - h}{h} \right)$$

Using LH rule

$$\lim_{h \rightarrow 0} \left(\frac{9}{4\pi} \right) \left(\frac{4^h \log 4 - 1}{1} \right)$$

$$\text{Taking limit } \left(\frac{9}{4\pi} \right) (\log 4 - 1)$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,B)

$$\text{Clearly } \lim_{x \rightarrow \alpha^+} \frac{\sin([x^3] - [x]^3)}{(x - \alpha)^3} = 0$$

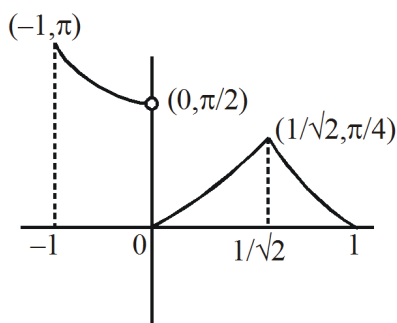
$\therefore$ for existence of limit L.H.L = 0

$$\Rightarrow \alpha^3 - 1 - (\alpha - 1)^3 = 0$$

$$\Rightarrow \alpha^3 - 1 - (\alpha^3 - 3\alpha^2 + 3\alpha - 1) = 0$$

$$3\alpha^2 - 3\alpha = 0 \Rightarrow \alpha = 0, 1$$

8. Ans (A,C,D)



$$\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 1$$

9. Ans (A,B,C,D)

$$y \sin \theta = x$$

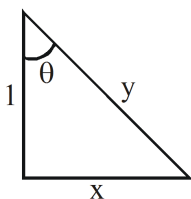
$$y \cos \theta = 1$$

$$(y^n - x^n) = \frac{1 - \sin^n \theta}{\cos^n \theta}$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin \theta}{\cos \theta} = 0,$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 \theta}{\cos^2 \theta} = 1$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^3 \theta}{\cos^3 \theta} = \infty \Rightarrow \text{Limit does not exist.}$$



10. Ans (A,B,C,D)

From the given expression $x^2 + y^2 + z^2 = xy + yz$

$$+ zx \Rightarrow a = b = c.$$

Hence all options are correct.

11. Ans (A,B)

$$\begin{aligned} \sum_{r=2}^{100} \frac{2^r (2-r)}{r(r+1)(r+2)} &= \sum_{r=2}^{100} \frac{2^r (r+2-2r)}{r(r+1)(r+2)} \\ &= \sum_{r=2}^{100} \frac{2^r}{r(r+1)} - \frac{2^{r+1}}{(r+1)(r+2)} = \frac{2}{3} - \frac{2^{101}}{(101)(102)} \end{aligned}$$

12. Ans (A,C)

$$\text{Let } \log_2 x = t$$

$$t^2 = t - t^2 + 1$$

$$2t^2 - t - 1 = 0$$

$$\Rightarrow (2t+1)(t-1) = 0 \Rightarrow t = -\frac{1}{2}, 1$$

$$x = \frac{1}{4}, 2 \left(x = \frac{1}{4} \text{ rejected} \right)$$

PART-3 : MATHEMATICS

SECTION-II

2. Ans (2.66 or 2.67)

$$\lim_{x \rightarrow -\infty} \sqrt{x^2 - \frac{4x}{3}} + x + 2$$

$$\lim_{x \rightarrow -\infty} \left(\frac{x^2 - \frac{4x}{3} - (x+2)^2}{\sqrt{x^2 - \frac{4x}{3}} - (x+2)} \right)$$

$$\lim_{x \rightarrow -\infty} \frac{-\frac{16}{3}x - 4}{|x| \sqrt{1 - \frac{4}{3x}} - (x+2)}$$

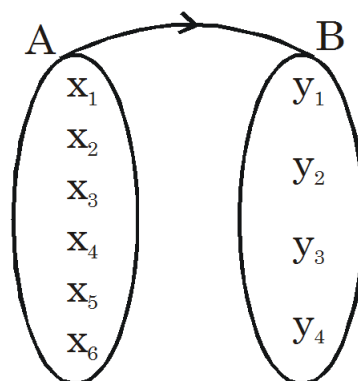
$$\lim_{x \rightarrow -\infty} \frac{x \left(-\frac{16}{3} - \frac{4}{x} \right)}{x \left[-\sqrt{1 - \frac{4}{3x}} - \left(1 + \frac{2}{x} \right) \right]}$$

$$\lim_{x \rightarrow -\infty} -\frac{16}{3(-2)} = \frac{8}{3}$$

3. Ans (0.50)

$$T_n = \tan^{-1} \left(\frac{\frac{n+1}{2} - \frac{n}{2}}{1 + \frac{n}{2} \cdot \frac{n+1}{2}} \right) = \tan^{-1} \left(\frac{n+1}{2} \right) - \tan^{-1} \frac{n}{2}$$

4. Ans (1560.00)



Groups 1, 1, 1, 3 and 1, 1, 2, 2

$$\frac{6! \times 4!}{3!(1!)^3 \times 3!} + \frac{6!(4!)}{(1!)^2 (2!)^2} \times \frac{1}{(2!)(2!)} = 1560$$

5. Ans (8.00)

$$\lim_{x \rightarrow 0} \left(\frac{f(x) - x + 2x^2 - 3x^3}{3x^4} \right) \cdot 2 = 4$$

$$\Rightarrow f(x) = x - 2x^2 + 3x^3 + 6x^4$$

$$\Rightarrow \sum_{i=0}^4 a_i = 1 - 2 + 3 + 6 = 8$$

6. Ans (4.00)

